

- 2 Body temperature and the degree of muscle contraction can be used to determine the time since death of a person.

The table shows how body temperature and body stiffness, due to muscle contraction, change with time since death.

| Time since death / hours | Body temperature | Body stiffness |
|--------------------------|------------------|----------------|
| < 3                      | warm             | not stiff      |
| 3 to 8                   | warm             | stiff          |
| 8 to 36                  | cold             | stiff          |
| > 36                     | cold             | not stiff      |

- (a) State how the temperature of a dead body should be measured.

(1)

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- (b) (i) Body temperature can be used to estimate the time since death using the following information:

- loss of  $0.78^{\circ}\text{C}$  per hour for the first 12 hours after death
- after 12 hours, loss of  $0.4^{\circ}\text{C}$  per hour.

Estimate the time since death of a person whose body temperature had fallen  $11.5^{\circ}\text{C}$ .

Give your answer to the nearest hour.

(2)

Answer ..... hours

(ii) Explain why this estimate would be different if the body had been left in a colder place.

(2)

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(c) Explain why using body stiffness only, as shown in the table, is insufficient to estimate the time since death accurately.

(3)

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**(Total for Question 2 = 8 marks)**